

Fin Plate Connection Design Report

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1 Input Parameters

Table 1: Input **Parameters**

	Value
Main Module	Connection
Module	Fin Plate Connection
Connectivity *	Column Flange-Beam Web
Axial (kN) *	40.0
Shear Force (kN) *	60.0

1.1 Supported Section - Mechanical Properties

Table 2: Section Details-Profile I Section

	Value
Designation	MB 500
Material	E 250 (Fe 410 W)B
Unit Weight (kN/m)	0.88
Cross Sectional Area, A (mm ²)	97.6
Depth, h (mm)	500
Width of Flange, bf (mm)	180
Web Thickness, tw (mm)	10.2

1.2 Supporting Section - Mechanical Properties

Table 3: Section Details-Profile I Section

	Value
Designation	UC 356 x 406 x 393
Material	E 300 (Fe 440)
Unit Weight (kN/m)	393
Cross Sectional Area, A (mm ²)	50,070
Depth, h (mm)	393.0
Width of Flange, bf (mm)	399.0

1.3 Fin Plate Details - Input and Design Preference

Table 4: Input **Parameters**

	Value
Thickness (mm) *	['3', '4', '5', '6', '8', '10', '12', '14', '16', '18', '20', '22', '24', '26', '28', '30']
Material *	E 165 (Fe 290)

1.4 Bolt Details - Input and Design Preference

Table 5: Input **Parameters**

	Value
Diameter (mm)	['16']
Grade *	['3.6', '4.6', '4.8', '5.6', '5.8', '6.8', '8.8', '9.8', '10.9', '12.9']
Bolt Type *	Bearing Bolt
Tension Type *	Pretensioned
Slip Factor *	0.3
Bolt Material Grade Overwrite	410
Bolt Hole Type	Standard

1.5 Detailing - Input and Design Preference

Table 6: Input **Parameters**

	Value
Corrosive Influences *	No
Edge Type *	Sheared or hand flame cut
Gap Between Beam End and Support Face (mm)	10

1.6 Weld Details

Table 7: Input **Parameters**

	Value
Weld Fabrication *	Shop Weld
Weld Material Grade Overwrite	410

2 Design Checks

2.1 Initial Section Check

Check	Required	Provided	Remarks
Shear Yielding of Supported Member (kN)	60.0	V _{dy} = 734.9 (kN)	Pass
Allowable Shear (kN)	60.0	V _d = 491.5 (kN)	Pass
Tension Yield Capacity of Member (kN)	40.0	T _{dy} = 918.6 (kN)	Pass

2.2 Load Consideration

Check	Required	Provided	Remarks
Applied Axial Force (kN)	40.0	40.0	
Applied Shear Load (kN)	60.0	V _{app} = 60.0, V _{min} = 60.0	

2.3 Bolt Design

	Check	Required	Provided	Remarks
Bolt Diameter (mm)			16	
Bolt Grade			8.8	
Minimum Plate Thickness (mm)	t_w = 10.2 mm		t_p = 10 mm	Fail
Number of Columns			n_c = 2	Pass
Number of Rows			n_r = 3	
Minimum Pitch (mm)	p_min = 48 mm		p = 50 mm	Pass
Minimum End Distance (mm)	e_min = 26 mm		e = 30 mm	Pass
Minimum Edge Distance (mm)	e_min = 26 mm		e = 35 mm	Pass

2.4 Section Design

	Check	Required	Provided	Remarks
Shear Capacity of Fin Plate (kN)		60.0	V_dp = 128.4 (kN)	Pass
Bearing Capacity of Plate (kN)		60.0	V_dpb = 96.2 (kN)	Pass
Block Shear Capacity (kN)		60.0	V_db = 152.7 (kN)	Pass
Moment Demand on Bolt Group (kNm)	M_e = 1.44 (kNm)		M_cap = 3.12 (kNm)	Pass

2.5 Weld Design

	Check	Required	Provided	Remarks
Weld Size (mm)	Min = 3, Max = 8.16		w = 6	Pass
Weld Strength (MPa)	f_u,w = 410		f_wd = 189.4	Pass